MILITARY SPECIFICATION

ADAPTER AND TEST PLUG ASSEMBLY: TRAINING

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions and the preparation for delivery criteria for one type of Adapter and Test Plug Assembly: Training.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-50293 - Plug, Test, Atomic Weapon
MIL-P-116 - Preservation, Methods of
MIL-S-60902(MU) - Soldering, Electrical Connections for Nuclear Weapons Systems, Procedure for

STANDARDS

MILITARY

MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipments of Inert Ammunition Components
MS16535-162 - Rivet Tublar
MS25281-F-10-Clamp
MS27183-4-Washer

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8850963 - Connector Plug Electrical
8861386 - Plug, Test, Atomic Weapon
8861390 - Connector for Plug Electrical
8883536 - Bag, Packing for Repair Parts
9227132 - Adapter, Cable to Connector; TNG
9227133 - Adapter and Test Plug Assembly
9229058 - Nut Coupling

FSC 1195

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9229059 - Tube Assembly
EL-9227133 - Equipment List

(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (dwg) 9227133, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Circuit resistance (see dwg 9227132). The circuit resistance shall comply with Table I when tested as specified in 4.4.1.

Table I
Circuit resistance

8861390 Pin	8850963 Pin	Ohms Max
C	C	0.1
B	B	0.1
E	E	0.1
A	A	0.1

3.5 Insulation resistance (see dwg 9227132). The insulation resistance of any circuit shall not be less than 20 meghoms when subjected to a potential of 500 ± 50 volts direct current (Vdc) with the current limited to 5 milliamperes (ma) when tested as specified in 4.4.2.

3.6 Vibration. The assembly shall withstand vibration when tested as specified in 4.4.3 without evidence of cracks, chips, looseness of parts, distortion or other deterioration. After the test, the assembly shall comply with 3.4 and 3.5.

3.7 High temperature. The assembly shall withstand a temperature of +125 ± 5 degrees fahrenheit ($^{\circ}$F) without evidence of swelling, peeling, cracking, distortion or other deterioration and shall comply with 3.4 and 3.5 when tested as specified in 4.4.4.

3.8 Low temperature. The assembly shall withstand a temperature of -13 ± 5°F without evidence of shrinkage, embrittlement, cracking, chipping, distortion or other deterioration and shall comply with 3.4 and 3.5 when tested as specified in 4.4.5.

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3.9 Humidity. The assembly shall withstand a minimum relative humidity of 90 percent without evidence of swelling, discoloration, peeling, cracking, distortion, shrinkage, corrosion, exposure of base material or other deterioration and shall comply with 3.4 and 3.5 when tested as specified in 4.4.6.

3.10 Workmanship

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.10.6 Electrical. Wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection

4.2.1 Quantity to be submitted (see 6.5). When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 9227133), 1 subassembly before potting (see dwgs 9227132 and 8861386) and 1 set of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table II. Tests shall be conducted in order specified in table II.

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Table II
First Article Inspection

Examination or test	Contractor's inspection permissable before Submission	Dest.	Non Dest.	Assembly	Sub-Ass	Part
1. Classification of characteristics (see 4.3.2.1)	X		X	X	X	X
2. Circuit resistance (see 3.4)	X		X	X		
3. Insulation resistance (see 3.5)	X		X	X		
4. Vibration (see 3.6)		X		X		
5. High temperature (see 3.7)	X		X	X		
6. Low temperature (see 3.8)	X		X	X		
7. Humidity (see 3.9)		X	X	X		

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	Over 1120	45
26-27	22	48-51	30	113-131	38		

Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	Over 274	22
12-13	11	23-27	15	55-76			

4.3.2.1.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Connector plug electrical (see dwg 8850963); nut coupling (see dwg 9229058); tube assembly (see dwg 9229059); rivet, tublar (see MS16535-162); washer (see MS27183-4); clamp (see MS25281-F-10) covering details of dwgs 9227132 and 9227133.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Objective evidence (see 6.2) available for part in accordance with standard or drawing as applicable.....Visual

Minor:
201. Workmanship (see 3.10).....Visual/Manual

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4.3.2.1.2 Adapter, cable to connector; TNG (see dwg 9227132 covering a detail of dwg 9227133).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Soldering in accordance with MIL-S-60902(MU)	Visual/Manual
102.	Potting in accordance with dwg 9227132	Visual
103.	Coupling nut turns freely	Manual
104.	All components present	Visual
Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.10)	Visual/Manual

4.3.2.1.3 Adapter and test plug assembly; TNG (see dwg 9227133).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	All components present	Visual
102.	Correct cord and clamp assembly	Visual/Manual
103.	Test plug (see dwg 8861386) in accordance with MIL-P-50293(MU) with exception of section 5 and paragraphs relating to section 5 in section 4	Visual
Minor:		
201.	Workmanship (see 3.10)	Visual

4.3.2.1.4 Preparation for delivery (see section 5).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual
Minor:		
201.	Workmanship (see 3.10)	Visual

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4.3.2.1.5 Bag, Packing, for Repair Parts (see dwg 8883536 covering dwg part number 8883536-4 and section 5.2).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Material (see 3.1)	Visual
102.	Length before packing, min	*Measure
103.	Width before packing, min	*Measure
104.	Objective evidence (see 6.2) available for heat seal test in accordance with MIL-P-116	Visual
105.	Properly packed	Visual
106.	No tear, split, or crack in bag	Visual

*Measure with suitable scale.

Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.10)	Visual/Manual

4.3.2.1.6 Packing and Marking for Storage and Shipments (see dwg 8808362 and section 5.2).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Gross load weight, max	*Measure
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Package material (see 3.1)	Visual

*Determine weight with a suitable scale.

Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.10)	Visual/Manual

4.3.3 Testing. All the tests specified in this paragraph shall be performed.

Test	Test Classification
a. Circuit resistance	Major
b. Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

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4.3.3.1 Circuit resistance (see 3.4). The circuit resistance shall be tested in accordance with 4.4.1. If the circuit resistance fails to comply with 3.4, the assembly shall be rejected and removed from the lot.

4.3.3.2 Insulation resistance (see 3.5). The insulation resistance of each adapter shall be tested as specified in 4.4.2. If any adapter fails to comply with 3.5, the adapter shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment List. (EL-9227133). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance (see 3.4). The circuit resistance between pins shall be tested with an approved electrical resistance tester in accordance with Table I.

4.4.2 Insulation resistance (see 3.5). The insulation resistance shall be tested with approved insulation resistance measuring equipment in accordance with Table V for 15 sec. minimum or until the testing device stabilizes.

Table V

Insulation Resistance (see dwg 9227132)

Connector 8861390 Pin	All pins and shells of connectors except Connector 8850963 Pin
C	C
B	B
E	E
A	A

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4.4.3 Vibration (see 3.6). The assembly shall be vibrated at a frequency of 5 to 500 to 5 c.p.s. With a constant double amplitude of 0.01 inch or an applied vibratory acceleration of 2 g's (in each direction) whichever is the limiting value. The rate of cycling shall be such that 15 minutes is required to proceed from 5 to 500 and back to 5 c.p.s., and the motion shall be applied for 2 hours in each plane. Resonant frequencies shall be noted and the assembly shall be vibrated at each resonant frequency noted for a period of 15 minutes. The vibrations shall be applied in the longitudinal plane and in a plane perpendicular to the longitudinal plane. After completion of test, the assembly shall comply with 3.4 and 3.5.

4.4.4 High temperature (see 3.7). The assembly shall be placed in a chamber and the temperature shall be raised to a temperature of 125 ± 5°F. After stabilization (see 6.4) the assembly shall be exposed to the temperature for 4 hours. The assembly shall then comply with 3.4 and 3.5.

4.4.5 Low temperature (see 3.8). The assembly shall be tested as specified in 4.4.4 except the temperature shall be lowered to -13 ± 5°F. After stabilization (see 6.4) the assembly shall be exposed to the temperature for 4 hours. The assembly shall then comply with 3.4 and 3.5.

4.4.6 Humidity (see 3.9). The assembly shall be placed in a humidity test chamber. Within a period of 1 hour, the test chamber temperature shall be raised to +125°F ± 5°F stabilized (see 6.4) and held for 2 hours. A relative humidity of 90 to 95 percent shall be maintained while the test chamber is +125°F ± 5°F. At the end of the 2 hour period, the chamber temperature shall be uniformly decreased during a 1 hour period, to between +68°F, and +95°F. During the period of decreasing temperature, the relative humidity shall be 95 + 5 percent. The assembly shall be subjected to ten such humidity cycles (1 hour of increasing temperature at 95 percent relative humidity, 2 hours of temperature stabilization at 90 to 95 percent relative humidity). The water used in controlling the relative humidity shall be distilled, having a pH value of between 6.5 and 7.5 + 77°F. Upon completion of the cycles, the assembly shall be immediately removed from the test chamber and dried by blowing dry, filtered, compressed air over the assembly for not more than 5 minutes. The assembly shall then comply with 3.4 and 3.5.

5. PREPARATION FOR DELIVERY

5.1 Production items

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

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5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1 of SQAP - Appendix I).

5.2.1.1 Level A. The adapter and test plug shall be packaged in accordance with dwgs 8883536 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The adapter and test plug shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The adapter and test plug shall be packed in accordance with dwgs 8883536 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwgs 8883536 and 8808362, as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Production or repair items.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table II) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table II) are verified by a representative designated by the Government.

Custodian:
Army - MU

Preparing Activity:
Army - MU
Project No. 1195-A061

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-A-50292(MU) Adapter and Test Plug Assembly

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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